EX. NO: 11 **DATE: 09/09/2026**

EXAMINE NETWORK ADDRESS TRANSLATION (NAT) USING CISCO PACKET TRACER

AIM:

To implement Network Address Translation (NAT) in Cisco Packet Tracer and allow private devices to communicate with an outside network.

PROCEDURE:

1) Add devices:

- 2 PCs
- 1 Switch
- 1 Router
- 1 Cloud / another Router

2) Connect devices:

- PC1 & PC2 → Switch
- Switch → Router G0/0
- Router G0/1 → Outside Router / Cloud

IP ADDRESS ASSIGNMENT:

Device	IP Address	Gateway
PC1	192.168.1.10	192.168.1.1
PC2	192.168.1.11	192.168.1.1
Router G0/0	192.168.1.1	-
Router G0/1	200.1.1.1	-

ROUTER CONFIGURATION:

Open Router → CLI and execute the following commands:

```
enable
configure terminal

! Interface Configuration
interface GigabitEthernet0/0
 ip address 192.168.1.1 255.255.255.0
```

```
 ip nat inside
 no shutdown
 exit

interface GigabitEthernet0/1
 ip address 200.1.1.1 255.255.255.0
 ip nat outside
 no shutdown
 exit

! Configure Static NAT
ip nat inside source static 192.168.1.10 200.1.1.10

! Configure Dynamic NAT
ip nat pool My-Pool 200.1.1.20 200.1.1.25 netmask 255.255.255.0
access-list 1 permit 192.168.1.0 0.0.0.255
ip nat inside source list 1 pool My-Pool
```

TEST NAT / VERIFICATION:

- From PC1, open Command Prompt and type: **`ping 200.1.1.2`**
- Check NAT translations on the router CLI: **`show ip nat translations`**

RESULT:

NAT was successfully configured & private IP addresses were translated to public IP addresses for communication with the outside network.